Plant leaf disease recognition based on ResNet fusion ECA

Yuehui Li[1,a]
[1]Harbin Institute of Information Technology
No. 9, University Town, Binxi Economic and Technological Development Zone, Harbin, Heilongjiang Province, China
[a]e-mail: hxciliyuehui@126.com

Xixiang Wang[2,b]
[2]Harbin University
109 Zhongxing Avenue, Harbin, Heilongjiang, China
[a]e-mail: wangxixiang930@sina.com

***Abstract*—In order to improve the performance of crop leaf detection, this experiment utilizes a large amount of image data of crop leaf diseases based on natural situations, and constructs a dataset of different crop leaf diseases in natural environments. These data are helpful for timely detection and identification of crop leaf diseases. Mainstream deep learning network models were used in this experiment for comparison and study, and the ECA channel attention mechanism module was added. The final experimental results show that the improved ResNet50_ ECA1 improves the accuracy of the original model by 1.1 percentage points loss value is reduced by 0.017, and ResNet50_ ECA1 improves the accuracy of the original model by 1.8 percentage points loss value is reduced by 0.0463.It can be proved that the introduction of the channel attention module can improve the feature extraction ability of the model, increase the recognition accuracy, and has new application value and ideas to provide new methods for plant protection and further improve grain yield.**

Keywords-Plant leaf disease detection; ResNet; ECA; attention mechanism

I. INTRODUCTION

With the continuous development of agriculture, the scale of crop production is also getting bigger and bigger, crops are not only related to the survival and development of human beings, but also play an important role in the economic development of a country. Timely and accurately grasp the type of crop diseases, the degree of damage and development trend, can lead to quickly taking effective management measures to reduce the economic losses caused by the disease on agricultural production, as well as avoiding the ecological environment pollution caused by blindly using drugs. This provides a basis for the scientific development of disease prevention and control strategies.[1] Despite advances in agricultural technology, crop leaf diseases are hard to avoid and continue to spread, particularly in large-scale cultivation. Symptoms of crop diseases mainly appear at the leaf position, and most farmers take measures by observing the degree of leaf disease and the type of infection.[2] Part of the detection and identification of crop pests and diseases is based on on-site diagnosis, relying on manual or semi-automatic methods, by agricultural experts or experienced farmers to carry out on-site observation and judgment, this method is simple and easy to implement, but the efficiency and timeliness need to be improved, the recognition of the results of the individual experience and technical level of the impact, can not fully meet the needs of the rapid development of modern agriculture.[3] Therefore, the identification of crop leaf diseases is not only important for agricultural production, but also helps farmers to monitor crop growth, predict yields, and identify pests and diseases. The shape, color, texture and other characteristics of crop leaves are complex and diverse, and traditional manual identification methods are often time-consuming and labor-intensive with low accuracy. In recent years, the development of plant leaf-based disease identification methods has been extremely rapid. Mana et al. extracted plant leaf phenotypic feature data, including leaf color features, shape features and texture features. Among them, the wavelet transform was used to extract the texture features of plant leaves, and the PCA principal component analysis was used to reduce the dimension of the extracted feature data. Next, a BP neural network model is established to classify and recognize plants. Experiments are conducted using different combinations of features and the recognition accuracy is up to 83.9%.[4] Jiaxin Guo et al. used K-means algorithm to segment wheat leaf lesion area, extracted color features and texture features of wheat lesion area, and constructed a dataset. Then the LM neural network wheat leaf disease recognition model was established and the input data were recognized. The recognition rate of wheat leaf disease based on color and texture features was 95.3%. The LM neural network algorithm can be used to quickly and accurately identify diseased leaves of wheat in small samples.[5] This paper mainly through the ResNet model within the residual unit and the convolution layer to join the ECA module after the experiment can improve the evaluation index, can effectively solve the problem of crop leaf recognition, has a good practical value.

II. METHOD

A. DenseNet model

DenseNet, known as Densely Connected Convolutional Networks, is a deep convolutional neural network composed of tightly connected layers.[6] The core design concept of this network lies in the fact that each layer is directly connected to all previous layers, so that each layer is able to directly utilize the feature maps of all previous layers, which greatly enhances the feature utilization efficiency and propagation. By doing so, the architecture effectively addresses the issue of gradient vanishing and achieves improved model performance. Additionally, the high reuse of features reduces the number of parameters require.

B. VGG16 model

VGG16 is a deep convolutional neural network consisting of 13 convolutional layers and 3 fully connected layers.[7] The network is characterized by a small size 3x3 convolutional kernel and a large depth network structure to enhance feature extraction. The design goal of VGG16 is to perform a full performance evaluation of a network with a large depth by using a smaller 3x3 convolutional filter, thus increasing the sensory field of the network while decreasing the number of parameters and reducing the risk of overfitting, and improving the performance of the network without increasing the computational burden.

C. GooleNet model

GoogLeNet is a deep convolutional neural network characterized by taking special convolutional layers and Inception modules.[8] The Inception module consists mainly of 1x1, 3x3 and 5x5 convolutional layers, and a maximum pooling layer. These layers process feature maps of different sizes separately, and then integrate the results of these processes to form a deeper feature map. In addition, the Inception module utilizes 1x1 convolutional layers for feature dimensionality reduction, effectively reducing the amount of computation.

D. ResNet18 model

ResNet18 is a deep learning model consisting of an 18-layer convolutional neural network with 16 convolutional layers and 2 fully connected layers.[9] The main feature of this network is the introduction of residual blocks, which create a constant mapping that can be used to solve the problem of gradient vanishing and gradient explosion in deep neural networks, thus enabling the training of deeper networks. The residual block can contain the primitive function as one of its elements, and this design allows ResNet18 to learn and extract image features better, resulting in better classification performance.

III. IMPROVING THE RESNET MODEL

A. ResNet50 model

ResNet50 is a further optimization of ResNet18, which has been improved in several aspects, including network depth, complexity, number of parameters, number of channels, and jump connections.ResNet50 consists of a 50-layer convolutional neural network and 16 residual blocks, which are deeper and more complex compared to ResNet18.The residual blocks of ResNet50 use a 1x1 convolutional layer, a 3x3 convolutional layer and a 1x1 convolutional layer. The main purpose of this design is to reduce the number of channels and thus the number of parameters by using a 1x1 convolutional layer. At the same time, using the 1x1 convolutional layer to combine features from different channels allows the network to better learn features at different scales, thus improving the classification performance.

B. ECANet model

The core of a convolutional neural network is the convolution operation, which fusion of spatial and channel dimensional features by means of localized receptive field, the representation of the network can be improved by modeling the interdependence between the convolutional feature channels. ECANet[10] focuses on improving SENet[11] by proposing an ECA module, which involves a local cross-channel interaction strategy. This module does not involve dimensionality reduction and instead adopts an adaptive selection of one-dimensional convolutional kernel sizes. This leads to enhanced performance. Additionally, the module only introduces a small number of parameters, yet achieves a significant performance gain. A model diagram of the ECA channel attention mechanism is shown in Figure 1.

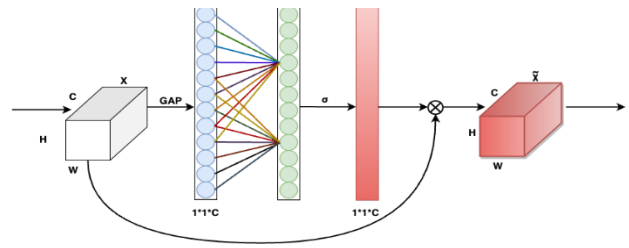

Figure 1 ECA model diagram

In the ECA module, the feature map was firstly global average Pooling (GAP), and then two Fully Connected (FC) layers with nonlinearity were used to connect, and a Sigmoid function was used to generate the channel weights. The original feature map and the corresponding elements of the feature channel weights were multiplied one by one to obtain the feature map with weights, so as to realize the focus of the model, ignore irrelevant information. σ is the Sigmoid function, C1D is the 1D convolution, g(X) is the global average pooling as shown in equation (1), and the feature channel weights are shown by the following equation (2).

$$g(x) = \frac{1}{WH}\sum_{i=1,j=1}^{W,H} x_{ij} \tag{1}$$

$$\omega = \sigma\left(C1D_K\left(g(x)\right)\right) \tag{2}$$

The coverage of cross-channel information interactions (i.e., the kernel size k of the 1D convolution) is proportional to the channel dimension C. There is a mapping relationship between k and C: K is the 1D convolution kernel size, where the value of K is as shown in equation (3).

$$K = \left|\frac{\log_2(C)}{\gamma} + \frac{b}{\gamma}\right|_{odd} \tag{3}$$

C. Improvement scheme based on ResNet50 fused ECA

The ECA module is added after the first layer of convolution to perform channel attention after the first large convolution (7*7 convolution kernel) of the model, this module is called ResNet50_ECA1, this improvement is shown in Figure 2.

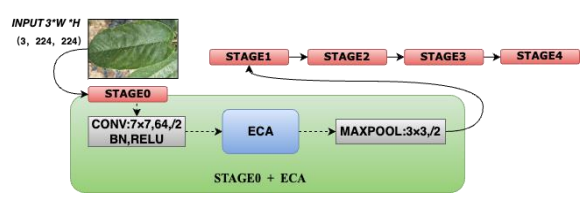

Figure 2 Diagram of the improved model of ResNet50 convolutional layer

The ECA channel attention module is added after the convolutional layer in the residual unit, and more attention modules are used to increase the capability of the model feature extraction, as shown in Figure 3 ECA module is added in the residual unit, and this improvement scheme is named ResNet50_ECA2.

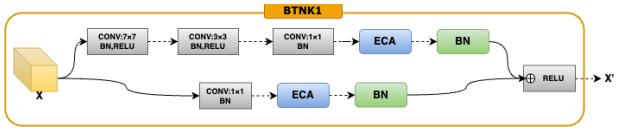

Figure 3 Residual unit added to ECA module

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental environment

The platform used in this experiment is Windows 11 operating system. the CPU is Intel(R) Xeon(R) CPU E5-2678 v3 with 2.50GHz and 32GB of RAM. The graphics card is NVIDIA GeForce RTX 3090 with 24G of GPU memory. the deep learning framework is PyTorch 2.0, CUDA version 11.1, and the interpreter is Python 3.9.

B. Data sets

This experiment was taken from the public dataset of Kaggle website, which contains a total of 154594 images of crop leaf diseases in JPG format, divided into 24 categories of crop leaf diseases, and partitioned the dataset according to the ratio of 8:1:1, dividing the dataset into 123676 training images, 15459 validation images and 15459 test images as shown in Table 1.

TABLE I. DISEASE TYPES AND ASSOCIATED PERCENTAGE SIZES

Index	Disease Name	Size	Relative scale(%)
0	Apple Apple scab	6049	3.913
1	Apple Black rot	5965	3.858
2	Apple Cedar apple rust	2641	1.708
3	Blight in corn Leaf	3439	2.225
4	Cherry (including sour) Powdery mildew	2527	1.635
5	Common Rust in corn Leaf	3919	2.535
6	Grape Black rot	11329	7.328
7	Grape Esca Black Measles	13285	8.593
8	Grape Leaf blight Isariopsis Leaf Spot	10333	6.684
9	Gray Leaf Spot in corn Leaf	1723	1.115
10	Orange Haunglongbing Citrus greening	52873	34.201
11	Pepper bell Bacterial spot	2989	1.933
12	Potato Late blight	3001	1.941
13	Strawberry Leaf scorch	2665	1.724
14	Tomato Bacterial spot	6382	4.128
15	Tomato Early blight	2461	1.592
16	Tomato Late blight	5728	3.705
17	Tomato Septoria leaf spot	5314	3.437
18	Tomato Spider mites Two spotted spider mite	5029	3.253
19	Tomato Target Spot	4213	2.725
20	Tomato Tomato mosaic virus	1120	0.724
21	algal leaf in tea	340	0.220
22	bird eye spot in tea	300	0.194
23	red leaf spot in tea	429	0.278

The data is uniformly 256*256, and the morphological processing of the images The processing process is as follows: first, binarization operation is performed on the images to obtain the binarized image, and then five corrosion operations are performed to eliminate the rest of the noises in the images, and then cropping is performed afterward to obtain the processed image that only retains the objects to be recognized. The processing flow is shown in Figure 4.

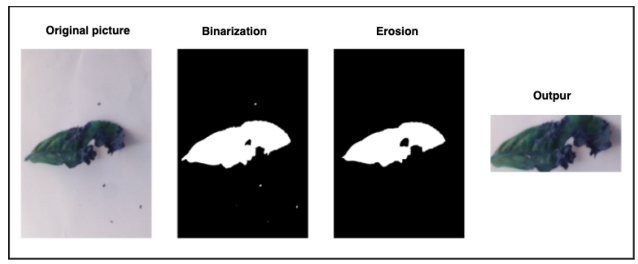

Figure 4 Image processing

C. Evaluation indicators

TP (True Positives): the number of positive classes predicted as positive; FN (False Negatives): the number of positive classes predicted as negative; FP (False Positives): the number of negative classes predicted as positive; TN (True Negatives): the number of negative classes predicted as negative. Accuracy is one of the common metrics used to evaluate the performance of a classification model, and it is used to measure the number of correctly classified samples of the model as a percentage of the total number of samples. The number of successfully classified samples is divided by the number of all samples. The advantage of accuracy is that it is easy to understand and calculate, and usually, the higher the accuracy, the better the performance of the classifier. The equation (4) for calculating the accuracy rate is as follows:

$$ACC = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{4}$$

Cross-Entropy loss function is a loss function commonly used to optimize model classification, which is widely used in deep learning. The Cross-Entropy loss function is generally used with e (natural number) as the base when dealing with multi-classification problems. The cross-entropy loss function is a convex function, so it can ensure that the optimization process converges to the global optimal solution, and secondly, the cross-entropy loss function is more sensitive to error, so it can update the model parameters faster, thus speeding up the training speed. The accuracy and generalization ability of the model can be improved by minimizing the cross-entropy loss function when training the deep neural network. The cross-entropy calculation equation (5) is as follows:

$$H(p,q) = -\sum p(x)\log\big(q(x)\big) \tag{5}$$

This experiment uses processed images as the dataset and different models are used for the experiment. The training loss as well as the accuracy of the ResNet50 model is better under the same data conditions.ResNet50 has a deep network structure and its network structure is shown in Table 2.

TABLE II. MODEL TRAINING DATA TABLE

Model	Train Loss	Train Acc(%)	Test Loss	Test Acc(%)
DenseNet121	0.407	98.6	0.065	97.8
VGG16	0.710	97.8	0.171	97.2
GoogleNet	0.224	92.3	0.200	93.6
ResNet18	0.084	97.0	0.110	96.4
ResNet50	0.047	98.9	0.045	97.9
ResNet50_ ECA1	0.003	99.1	0.015	99.0
ResNet50_ ECA2	0.0007	99.8	0.014	99.7

By table 1 we can see that the number of parameters of VGG16, which also has a deep network structure, is much higher than that of ResNet50, and the inference time of VGG is long and the weight of the model is heavy. Combining the recognition accuracy of the model and the number of parameters we choose to improve ResNet50, and the training loss and recognition accuracy of the two improved methods are improved compared with the native model. The training curve is shown in Figure 5.

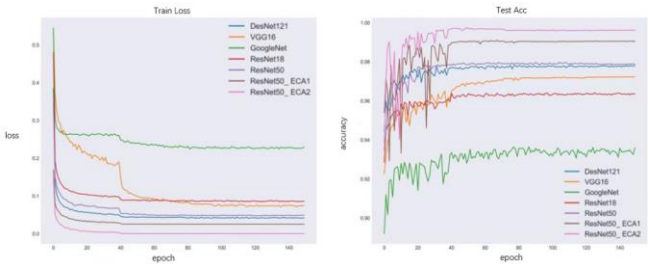

Figure 5 Model training curve graph

V. CONCLUSIONS

In this paper, a deep learning algorithm is used to classify and diagnose diseases on plant leaves, and by improving the ResNet50 model, the feature extraction effect and detection and classification ability of the model in processing plant leaf images are improved, as well as the accuracy and robustness of the model. This paper details two improvement schemes for the benchmark model ResNet50 to improve the accuracy of the model. This paper also uses image processing methods to process larger size images thereby retaining more information. The enhancement of the model by adding the attention module is verified through comparison experiments and group ablation experiments. The experimental results show that the improved model achieves 99.7% accuracy on the test set. In this paper, the improved model is evaluated and analyzed, and it is found that the improved model has better feature extraction and classification capabilities, as well as higher accuracy and robustness, which provides an automated plant disease detection method for agricultural production. Therefore, the improved ResNet50-based model proposed in this paper has high practical value and popularization value.

REFERENCES

[1] Zheng Guo,Jiang Yusong. Multi-task-based learning for crop leaf disease diagnosis [J]. China Agricultural Science and Technology Guide, 2024, 26 (01): 89-98. DOI:10.13304/j.nykjdb.2022.0650.

[2] Sun Yani. Research on image recognition technology of leaf diseases of typical crops in northern Shaanxi [D]. Yan'an University, 2023. DOI:10.27438/d.cnki.gyadu.2023.000773.

[3] Yue Xishen. Research progress of deep learning in crop disease recognition [J]. Anhui Agricultural Bulletin, 2024, 30 (06): 100-103. DOI:10.16377/j.cnki.issn1007-7731.2024.06.027.

[4] Ma Na,Ren Y Xiang. A study on multi-feature based BP neural network multiple plant leaf disease recognition[J]. Chinese Agronomy Bulletin, 2024, 40(4):158-164.DOI:10.11924/j.issn.1000-6850.casb2022-1025.

[5] MA Na,GUO Jiaxin. Recognition of leaf diseases in wheat based on LM neural network[J]. Journal of Agronomy, 2023, 13(2):8. DOI:10.11923/j.issn.2095-4050.cjas2022-0031.

[6] Huang G , Liu Z , Laurens V D M ,et al. Densely Connected Convolutional Networks[J].IEEE Computer Society, 2016.DOI:10.1109/CVPR.2017.243.

[7] Simonyan K , Zisserman A .Very Deep Convolutional Networks for Large-Scale Image Recognition[J].Computer Science, 2014.DOI:10.48550/arXiv.1409. 1556.

[8] Szegedy C , Liu W , Jia Y ,et al. Going Deeper with Convolutions[J].IEEE Computer Society, 2014.DOI:10.1109/CVPR.2015.7298594.

[9] Gao Y , Li K , Mosalam K M ,et al.Deep Residual Net with Transfer Learning for Image-based Structural Damage Recognition Deep Residual Network with Transfer Learning[C]//11th National Conference on Earthquake Engineering.2018.DOI:10.13140/RG.2.2.22764.26241.

[10] Wang Q , Wu B , Zhu P ,et al. ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks[C]//2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).IEEE, 2020.DOI:10.1109/CVPR42600.2020.01155.

[11] Hu J , Shen L , Sun G .Squeeze-and-Excitation Networks[C]//2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).IEEE, 2018.DOI. 10.1109/cvpr.2018.00745